

Research Interests

My research interests include Virtual and Augmented Reality, Human-Computer Interaction, and Visual Perception. My dissertation focuses on the design and evaluation of cybersickness mitigation techniques.

Education

- 2020 – 2026 **Ph.D. in Computer Science**, *University of Minnesota*, Minneapolis, MN, USA.
(Expected) Advisor: Evan Suma Rosenberg. Date of candidacy: Jan 3, 2023.
- 2019 – 2025 **M.Sc. in Computer Science**, *University of Minnesota*, Minneapolis, MN, USA.
- 2014 – 2018 **B.Eng. in Information Security**, *Harbin Institute of Technology*, Weihai, Shandong, China.
- 2017 **Visiting International Student**, *University of Wisconsin-Madison*, Madison, WI, USA.

Awards and Honors

- 2025 Doctoral Dissertation Fellowship, University of Minnesota
- 2025 Best Paper Award (top 1% of total submissions), IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

Publications

Journal Papers

- 2026 Tongyu Nie, Yuxuan Huang, Ville Cantory, Danhua Zhang, and Evan Suma Rosenberg. Effects of field-of-view restriction and peripheral teleportation on path integration during virtual locomotion. *IEEE Transactions on Visualization and Computer Graphics*, volume 32, pages 1–10, 2026. (Proc. IEEE ISMAR 2026).
- 2026 Tongyu Nie*, Ville Cantory*, Courtney Hutton Pospick*, Danhua Zhang, Haoyu Tan, Sam Adeniyi, Fei Wu, Daniel Zielasko, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Revenge of the Sick: A Meta-Analysis of Washout Periods in Cybersickness Research. *IEEE Transactions on Visualization and Computer Graphics*, volume 32, pages 1–20, 2026.
- 2025 Tongyu Nie, Courtney Hutton Pospick, Ville Cantory, Danhua Zhang, Jasmine Joyce DeGuzman, Victoria Interrante, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness During Virtual Locomotion. *IEEE Transactions on Visualization and Computer Graphics*, volume 31, pages 2891–2900, 2025. (Proc. IEEE VR 2025).

Conference Papers

- 2026 Xianhao Carton Liu, Difan Jia, Tongyu Nie, Evan Suma Rosenberg, Victoria Interrante, and Chen Zhu-Tian. Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision-Making? A Comparative Study of AR X-Ray vs. 2D Minimap. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI '26. Association for Computing Machinery, 2026.

- 2025 Jasmine Joyce DeGuzman, Kaori Hirano, Tabitha Peck, Alice Guth, Evan Suma Rosenberg, and Tongyu Nie. Reduction of Motion Complexity as an Objective Indicator of Cybersickness in Virtual Reality. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 156–164, 2025.
- 2025 Ville Cantory, Darya Biparva, Haoyu Tan, Tongyu Nie, John Schroeder, Ruofei Du, Victoria Interrante, and Piotr Didyk. Enhancing Foveated Rendering with Weighted Reservoir Sampling. In *The 18th ACM SIGGRAPH Conference on Motion, Interaction, and Games (MIG '25)*, pages 2–12, 2025.
- 2024 Haoyu Tan, Tongyu Nie, and Evan Suma Rosenberg. Invisible Mesh: Effects of X-Ray Vision Metaphors on Depth Perception in Optical-See-Through Augmented Reality. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 376–386, 2024.
- 2023 Tongyu Nie, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Like a Rolling Stone: Effects of Space Deformation During Linear Acceleration on Slope Perception and Cybersickness. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 658–668, 2023.

Extended Abstracts

- 2024 Jasmine Joyce DeGuzman, Erik DeVries Smith, Samyok Nepal, Kalinda Miller, Courtney Hutton Pospick, Tongyu Nie, and Evan Suma Rosenberg. Walk Me Through It: Using Impossible Spaces to Embody Graph Traversal Algorithms. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1092–1093, 2024.
- 2024 Catherine Borisova, Ruth Mesfin, Tongyu Nie, and Evan Suma Rosenberg. Build Your Own Trigonometric Adventure in Virtual Reality. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1094–1095, 2024.
- 2021 Tongyu Nie and Evan Suma Rosenberg. Redirected Tilting: Eliciting Postural Changes with a Rotational Self-Motion Illusion. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 178–182, 2021.

Under Review

- 2026 Danhua Zhang, Evan Suma Rosenberg, Pan Hao, Ville Cantory, Fei Wu, Ziyuan Tang, Haoyu Tan, Tongyu Nie, Qiuxuan Wu, and Ruofei Du. Lowfrictionagent: Low-friction human-agent configuration and real-time multimodal interaction in vr. In *ACM Symposium on User Interface Software and Technology (Under Review)*, 2026.
- 2026 Danhua Zhang, Cynthia Sherraden Bradley, Christopher Kissell, Courtney Hutton Pospick, Seraphina Yong, Tongyu Nie, and Evan Suma Rosenberg. Iterative design and evaluation of a vr nursing training system: Insights from clinicians and student learners. *Virtual Reality (Under Review)*, 2026.
- 2026 Yuxuan Huang, Qiao Jin, Tongyu Nie, Victoria Interrante, and Evan Suma Rosenberg. Secure text entry using a virtual radial keyboard with dynamically resized keys and non-intrusive randomization. In *ACM Symposium on User Interface Software and Technology (Under Review)*, 2026.
- 2026 Yuxuan Huang, Qiao Jin, Tongyu Nie, Victoria Interrante, and Evan Suma Rosenberg. Context when needed, digits when useful: Balancing memorability and efficiency in virtual reality authentication. In *IEEE International Symposium on Mixed and Augmented Reality (Under Review)*, 2026.

Research Experience

- 08/2020 – **Graduate Research Assistant**, *Illusioneering Lab*, University of Minnesota.
- Present
- Developed two novel cybersickness mitigation techniques using shaders in HLSL in collaboration with multi-disciplinary researchers in Kinesiology.
 - Explored the use of geometry deformation on cybersickness and uneven terrain simulation.
 - Designed a rest frame using content from the same virtual environment users immersed in and achieved state-of-the-art results in cybersickness mitigation.
 - Evaluated the impact of mitigation techniques on visual path integration.
 - Conducted a meta-analysis to provide guidance for cybersickness experimental design.
 - Mentored multiple undergraduate and graduate students, leading to publications in premier venues.
- Summer 2024–2026 **NSF REU Mentor**, *REU Site: Human-Centered Computing for Social Good*, University of Minnesota.
- Supervised 10+ undergraduate students from underrepresented groups during their summer research experiences; served as a point of contact and intermediary between students, the department, and supervising professors.
 - Helped students on individual projects in graphics, visualization, virtual reality, robotics, and human-computer interaction; wrote and taught 8 weekly modules covering the research process, data analysis, and scientific writing.
 - Organized a graduate student panel on graduate school pathways.
- 05 – 08/2021 **Research Intern**, *Institute of Creative Technology*, University of Southern California.
- Designed novel uncertainty visualization metaphors for augmented reality display.
 - Conducted an online human-subject study to evaluate the effectiveness of visualization techniques.

Mentoring Experience

Master's Students

2022 – 2024 Haoyu Tan (Now: PhD at UMN)

Undergraduate Students

2025 Meryem Koksall

2021 – 2025 Jasmine Joyce DeGuzeman (Now: PhD at UCF)

2024 Kaori Hirano

2024 Shauntavia Hooper

2023 – 2024 Catherine Borisova

2023 – 2024 Ruth Mesfin

2022 Christian Halvorson

2022 Caleb Calomarde

Teaching Experience

Graduate Teaching Assistant, *University of Minnesota*, Duties included leading labs, holding office hours, grading, and developing course materials.

Spring, 2025 CSCI 1933: Introduction to Algorithms and Data Structures

Fall, 2024 CSCI 1933: Introduction to Algorithms and Data Structures

Fall, 2023 CSCI 5619: Virtual Reality and 3D Interaction

Spring, 2022 CSCI 5552: Sensing and Estimation in Robotics

Fall, 2021 CSCI 1913: Introduction to Algorithms, Data Structures, and Program Development

Spring, 2021 CSCI 1913: Introduction to Algorithms, Data Structures, and Program Development

Fall, 2020 CSCI 5619: Virtual Reality and 3D Interaction

Academic Activities

Workshop Organization

2026 IEEE VR Workshop on Immersive Sickness Prevention (WISP) Co-organizer

Program Committee Service

2025 – 2026 EuroXR Scientific International Program Committee
2026 IEEE ISMAR 2026 Poster Program Committee
2026 IEEE VR 2026 Poster Program Committee
2025 IEEE VR 2025 Poster Program Committee
2021 – 2025 IEEE VR Workshop on Immersive Sickness Prevention (WISP) Program Committee

Journal & Conference Reviewing

2023 – 2026	IEEE Conference on Virtual Reality and 3D User Interfaces (VR)	21 reviews
2023 – 2026	IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	20 reviews
2024 – 2025	IEEE Visualization Conference (VIS)	3 reviews
2025	IEEE Pacific Visualization Conference (PacificVis)	2 reviews
2024 – 2026	ACM CHI Conference on Human Factors in Computing Systems (CHI)	9 reviews
2026	ACM Symposium on User Interface Software and Technology (UIST)	1 review
2025	ACM Conference on Computer-Supported Cooperative Work (CSCW)	1 review
2023 – 2026	ACM Symposium on Virtual Reality Software and Technology (VRST)	9 reviews
2024 – 2025	ACM Symposium on Spatial User Interaction (SUI)	5 reviews
2023 – 2026	IEEE Transactions on Visualization and Computer Graphics (TVCG)	5 reviews

Academic Service

Student Volunteer.

IEEE VR 2020, IEEE VR 2024, IEEE ISMAR 2024

University/Department Service.

2024 Human Centered Computing Seminar: Student Coordinator
2022 Council of Graduate Students Grants Review Committee: Reviewer
2021 Empowering Women in Science – A Friend in STEM Mentoring Program: Mentor

Invited Presentation

Invited Talk.

- Extended Horizon Lab Meeting Invited Talk. “Design and Evaluation of Cybersickness Mitigation Techniques in Virtual Reality” North Carolina State University (Raleigh, NC). Feb 17, 2026.
- Workshop on Immersive Sickness Prevention. “The Influence of Carryover Effects on Cybersickness Experiment Outcomes: A Meta-Analysis” IEEE VR 2025 (Saint-Malo, France). Mar 8, 2025.
- Workshop on Locomotion and Wayfinding in XR. “Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness During Virtual Locomotion.” IEEE VR 2025 (Saint-Malo, France). Mar 8, 2025.
- Virtual Experiences Research Group Lab Meeting Invited Talk. “Advancing Cybersickness Mitigation in Virtual Reality” University of Florida (Gainesville, FL). Feb 17, 2025.
- Workshop on Immersive Sickness Prevention. “Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness.” IEEE VR 2024 (Orlando, FL). Mar 16, 2024.

Guest Lecture.

- CSCI 5619: VR and 3D Interaction “Cybersickness in Virtual Reality.” University of Minnesota. Nov 14, 2025.
- CSC 591 (008): XR and 3D Interaction “Cybersickness in Virtual Reality.” North Carolina State University. Oct 30, 2025.
- CSCI 8001: Intro to Research in CS, I “Grad Student Panel.” University of Minnesota. Sept 19, 2025.
- CSCI 1933: Intro to Algorithms & Data Structures “Exceptions and Iterators in Java.” University of Minnesota. March 28, 2025
- CSCI 5619: VR and 3D Interaction “Motion Filtering.” University of Minnesota. Oct 13, 2023.
- CSCI 1913: Intro to Algorithms & Program Development “Version Control System and Git.” University of Minnesota. Apr 15, 2021.

Referees

Dr. Evan Suma Rosenberg

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